

Uttarakhand Electricity Regulatory Commission (Distribution Code) Regulations, 2018

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State of Uttarakhand - Act

Uttarakhand Electricity Regulatory Commission (Distribution Code) Regulations, 2018

UTTARAKHAND

India

Uttarakhand Electricity Regulatory Commission (Distribution Code) Regulations, 2018

Rule UTTARAKHAND-ELECTRICITY-REGULATORY-COMMISSION-DISTRIBUTION-CODE-REGULATIONS-2018 of 2018

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Uttarakhand Electricity Regulatory Commission (Distribution Code) Regulations, 2018

Published vide Notification No. F-(9)27/RG/UERC/2018/2016, dated 31.3.2018

Notification No. F-(9)27/RG/UERC/2018/2016. - In exercise of the powers conferred on it by Section 181 of the Electricity Act 2003, read with clause (c), (e) and (i) of sub-section (1) of Section 86 of the said Act and Clause 18 of Distribution and Retail Supply Licence issued to the Distribution Licensee under Section 14 of the Act and all powers enabling it in that behalf, the Uttarakhand Electricity Regulatory Commission hereby makes the following regulations:

Chapter 1

General

1.

1. Short title, commencement and interpretation.

(1)

These regulations may be called the Uttarakhand Electricity Regulatory Commission (Distribution Code) Regulations, 2018.

(2)

This regulation shall be applicable to all Distribution System participants including:

(a)

Distribution Licensee(s) [including deemed licensee(s)];

(b)

Open Access Customers connected to Distribution Systems;

(c)

Other Distribution Licensee connected to Distribution Systems;

(d)

Institutions covered under Proviso 8 of Section 14 of the Act;

(e)

Embedded Generators; and

(f)

Consumers.

(3)

These regulations shall come into force on the date of the publication in the Official Gazette replacing "Uttarakhand Electricity Regulatory Commission (Distribution Code) Regulations, 2007" thereof.

(4)

These regulations shall be interpreted and implemented in accordance with, and not at variance from, the provisions of the Electricity Act, 2003 read with the Central Electricity Authority (Measures relating to Safety and Electric Supply) Regulations, 2010 as amended from time to time and any other regulations issued by Central Electricity Authority in this regard.

1.2

Definitions. - (1) In the Distribution Code the following words and expressions shall, unless the subject matter or context otherwise requires or is inconsistent therewith, bear the following meanings:

(a)

"Act" means The Electricity Act, 2003 (Act 36 of 2003).

(b)

"Agreement" means an agreement entered into by the Distribution Licensee and the User.

(c)

"Apparatus" means electrical apparatus and includes all machines, fittings, accessories and appliances connected to the Electrical Distribution System.

(d)

"Area of Supply" means the geographic area within which Licensee is for the time being authorised by his licence to supply electrical energy.

(e)

"CBIP" means Central Board of Irrigation and Power.

(f)

"CEA" means Central Electricity Authority.

(g)

"Circuit" means an arrangement of conductor(s) for the purpose of conveying electrical energy and forming a system or a branch of system.

(h)

"Commission" means Uttarakhand Electricity Regulatory Commission.

(i)

"Conductor" means any wire, cable, bar, tube, rail or plate used for conducting energy and electrically connected to the

system.

(j)

"Connected Load" means aggregate of the manufacture's rating of all energy consuming apparatus duly wired and connected to the power supply system of licensee including portable apparatus in the consumer's premises. This shall not include the load of spare plug, sockets, load exclusively installed for fire fighting purposes. The load of either water and room heating or room cooling apparatus, whichever is higher, shall be taken into account as per prevailing season (1st April to 30th September for cooling use and 1st October to 31st March for heating use).

The definition of connected load shall be used only for the purpose of assessment in case of direct theft or dishonest abstraction of energy or unauthorised use of energy.

(k)

"Control Person" means a person identified as having technical capability and responsibility for cross boundary safety.

(l)

"OCR" means Distribution Code Review.

(m)

"DCRP" means Distribution Code Review Panel.

(n)

"Embedded" means having a direct electrical connection to an intra state electrical system.

(o)

"Extra High Tension (EHT) or Extra High Voltage (EHV)" means where the voltage exceeds 33,000 volts under normal conditions, subject to the percentage variation permissible.

(p)

"GSS" means Grid Sub-station.

(q)

"Harmonics" means a component of a periodic wave having frequency that is an integral multiple of the fundamental power line frequency of 50 Hz causing distortion to pure sinusoidal wave form of voltage or current, and as governed by IEEE STD 519-1992, namely "IEEE Recommended Practices and Requirements for Harmonic Control in Electrical Power Systems" and CEA Safety Regulations/ Grid Standards as amended from time to time.

(r)

"High Tension (HT) or High Voltage (HV)" means the voltage between 650 volts and up to 33,000 Volts under normal conditions, subject to the percentage variation permissible.

(s)

"Indian Standards (IS)" means those Standards and specifications approved by the Bureau of Indian Standards.

(t)

"Interface Point"/"Connection Point"/Boundary Point" means point at which a User's electrical system is connected to the Licensee's Distribution System.

(u)

"Low Tension (LT) or Low Voltage (LV)" means the voltage of 230 volts between phase and neutral or 400 volts between any two phases under normal conditions, subject to the percentage variation permissible.

(v)

"Power Factor" means the ratio of Active Power (kW) to Apparent Power (KVA),

(w)

"PTW" means permit to work.

(x)

"REC" means Rural Electrification Corporation.

(y)

"Transmission System" means the system consisting of extra high voltage electric lines being operated at EHT (excluding generator interconnection facilities) owned and or operated by the Transmission Licensee for the purposes of the transmission of electricity from one power station to a sub- station or to another power station or between sub-stations or to or from any external interconnection equipment up to the interconnection with the distribution system, any plant and apparatus and meters owned or used by the Transmission Licensee in connection with the transmission of electricity, but shall not include any part of the Licensee distribution system.

(z)

"User" means any person having electrical interface with, or using the Distribution System of the Distribution Licensee to whom this Code is applicable. Any other Distribution Licensee, Transmission Licensee and generating units connected to the Distribution system are also included in this term.

(aa)

All words and expressions used and not defined in these regulations but defined in The Electricity Act, 2003 shall have the meanings as assigned to them in the said Act.

1.3

Objectives. - (1) To ensure that distribution system is developed and maintained in an efficient, coordinated and economical manner and the Distribution Licensee and all Distribution System Users comply with respective obligations as specified in the Act.

(2)

The Distribution Code brings together a single set of regulations, for using the Distribution network and provides the following:

(a)

Technical aspects of working relationship between the Licensee's Distribution system and to those connected and seeking connection to it.

(b)

Facilitation of the operation, maintenance, development and planning of economic and reliable power distribution network.

(c)

Safety and Protection parameters to be followed by the distribution licensee and users.

1.4

Scope of Distribution Code. - (1) The Distribution Code shall cover all material technical aspects relating to connections to and the operation and use of the Distribution System including the operation of the electrical lines and electrical plant and apparatus connected to the Distribution System insofar as relevant to the operation and use of the Distribution System and shall include:

(a)

Distribution planning and connection code containing connection conditions specifying the technical, design and operational criteria to be complied with by any person connected or seeking connection with the Licensee's Distribution System and planning codes specifying the plan required for laying the distribution lines and the services lines in the Area of Supply, the technical and design criteria and procedures to be applied by the Distribution Licensee in the planning and development of the Licensee's Distribution System; and

(b)

A distribution operation code specifying the conditions under which the Distribution Licensee shall operate its Distribution System and under which persons shall operate their plant and/or Distribution System in relation to the Licensee's Distribution System, insofar as necessary to protect the security and quality of supply and safe operation of the Licensee's Distribution System under both normal and abnormal operating conditions.

(2)

The Distribution Code is not exhaustive as to the requirements to be complied with by the Distribution Licensee and the Users connected or seeking connection to the Licensee's Distribution System. The Distribution Licensee and all Users/Consumers must also comply with the requirement as laid down in various codes, standard and regulations under relevant laws in force.

(3)

The Distribution Code also deals in terms of distribution management in the event of outages and shortages of Electricity Supply and distribution thereof amongst all the categories of consumers as per the system network requirement. However, consumers having captive power plants shall come to the rescue of the Distribution licensee as first priority in the event of outages and shortages and resort to load shedding immediately on instructions from the Distribution licensee/SLDC.

(4)

This Distribution Code shall include-

- (a)
Management of Distribution Code
- (b)
Distribution System Planning Code
- (c)
Connectivity Conditions
- (d)
Operation Code
- (e)
Distribution Protection Requirement
- (f)
Cross Boundary Safety Code
- (g)
Incident/Accident Reporting

1.5

Implementation and Operation of the Distribution Code. - (1) The Distribution Licensee shall be responsible for its implementation within its Area of Supply. Users shall comply with the provisions of this code.

(2)

If any User has any difficulty in complying with any of the provisions of the Distribution Code, he shall immediately, inform to the Distribution Licensee and/or Commission, as the case may be.

(3)

Any continued non-compliance, without reasonable grounds shall constitute a deviation under the Act, and may lead to disconnection of the User's plant or apparatus in line with the provisions of The Electricity Act, 2003 from the Licensee's Distribution System. The responsibility for the consequences of disconnection including payment of damages and others rests with the User(s) who consistently violates the Distribution Code.

(4)

Non-compliance with any provision of the Distribution Code by the Distribution Licensee shall attract the consequences as provided in the Acts or in the Licence. However, in the event of non-compliance with the Distribution Code, the Distribution Licensee shall prepare and submit to the Commission an action plan for ensuring the compliance of the provisions of the Distribution Code. The Commission may exempt Distribution Licensee from compliance of any provisions for a particular period, if it is found that the compliance is not feasible for such period, considering the resources available and the circumstances prevailing.

1.6

Limitations of the Distribution Code. - (1) Nothing contained in this Code should be interpreted as imposing obligations/duties on Consumers/Distribution Licensees greater or more onerous than those mentioned in the Electricity Act, 2003 under relevant sections.

(2)

The Distribution Code contains procedures for the management of day to day technical situations in the Distribution System, taking into account a wide range of operational conditions likely to be encountered under both normal and abnormal conditions. The Distribution Code cannot foresee all the possible operating conditions. Users must therefore understand and accept that the Distribution Licensee, in such unforeseen circumstances, may be required to act decisively and with due expedition to discharge his obligations under the Licence. Users shall provide such reasonable co-operation and assistance as the Distribution Licensee may require in such circumstances. The concerned Distribution Licensee shall however refer all such cases for ratification in the next meeting of the Distribution Code Review Panel as described under Chapter 2 of the code 'Management of Distribution Code'.

1.7

Confidentiality. - Under the terms of Distribution Code, the Distribution Licensee will receive information from Users relating to their business. The Distribution Licensee shall not, other than as required by Distribution Code, disclose such information to any other person without the prior written consent of such informant, unless required by Central/State government departments or the authority.

1.8

Procedures to settle disputes. - In the event of any dispute regarding interpretation of any regulations provided in the Distribution Code between any User and the Distribution Licensee, the matter shall be referred to the Distribution Code Review Panel and thereafter to Uttarakhand Electricity Regulatory Commission. Commission's decision shall be final and binding on both the parties.

Chapter 2

Management of Distribution Code

2.

1. Introduction.

- This chapter defines the method of managing Distribution Code, pursuing of any changes/modifications required and the responsibilities of the Distribution Licensees and the Users in this regard. This section facilitates revisions taking into account the views of all parties in an equitable manner.

2.2

Objective. - The objective of this chapter is to define the functions of the Distribution Code Review Panel and procedure for review and revision of the Distribution Code.

2.3

Distribution Code Review Panel. - (1) A standing body called Distribution Code Review Panel (DCRP) shall be constituted by the Commission comprising of the representatives of Distribution Licensee as well as the Users of the Distribution System in line with the provisions of this Code.

(2)

No change in this Distribution Code, shall be made without being deliberated upon and agreed to by the Distribution Code Review Panel and thereafter approved by Commission. However, in an unusual situation where normal day-to-day operation is not possible without revision in some of the provisions of the regulations of Distribution Code, a provisional revision may be implemented before approval of Commission is received, but only after discussion at a special Review Panel Meeting convened on emergency basis. Commission should promptly be intimated about the provisional revision. Commission may issue directions required to revise the Distribution Code accordingly as may be specified in those directions and the Distribution Licensee shall promptly comply with any such directions.

(3)

The Distribution Code Review Panel shall be composed of the following members, who shall be notified by the Commission.

(a)

Director (Technical/Operation) of Concerned Distribution Licensee(s);

(b)

Chief Engineer/General Manager level officer from other Distribution Licensee(s) in the State;

(c)

Chief Engineer/General Manager Level officer from STU;

(d)

One member not below the rank of Superintending Engineer/D.G.M. to be nominated by the SLDC;

(e)

One member not below the rank of Chief Engineer/G.M. representing State owned Generating Company;

(f)

One member not below the rank of Chief Engineer/G.M. representing other Generating Companies in the state;

(g)

One member not below the rank of Chief Project Officer representing State Renewable Energy Agency;

(h)

One member representing open access consumers;

(i)

One member representing Industrial Consumers;

(j)

One member representing Domestic/Commercial Consumers;

(k)

One member representing Agriculture Consumers;

(l)

One member representing RE generators in the State;

(m)

One member of State Advisory Committee nominated by the Commission.

2.4

Terms of Office. - The Chairman of the Distribution Code Review Panel shall be Director (Technical/Operation) of Distribution Licensee. The Distribution Code Review Panel shall however be perpetual under the Distribution Code. All members of the Distribution Code Review Panel as mentioned at Sl. No. (a) to (g) shall hold office until changed/replaced by the respective parent organisation. All members of the Distribution Code Review Panel as mentioned at Sl. No. (h) to (l) shall hold office in rotation among all such organisations in the State. The term of each such member shall be two (2) years. However, the Commission shall have the powers to increase or decrease the aforementioned tenure.

2.5

DCRP Support Staff and Operating Cost. - The Member from Distribution Licensee holding the office of Chairman of DCRP at a given time shall provide secretarial staff including secretary not below the rank of S.E./D.GM. required to support DCRP operations. The cost associated with such secretarial assistance shall also be borne by that Distribution Licensee.

2.6

Functions of the Review Panel. - The functions of the Review Panel shall be:

(1)

Maintenance of the Distribution Code and its working under continuous scrutiny and review.

(2)

Consideration of requests for review made by any user and publication of their recommendations for changes in the Distribution Code together with reasons for such changes.

(3)

Provide guidance on interpretation and implementation of the Distribution Code.

(4)

Examination of the problems raised by any User as well as resolution of the problems.

(5)

Ensuring that the changes/modifications proposed in the Distribution Code are consistent and compatible with standard technical manual or guidelines, codes, laws, acts, rules and regulations in force at that point of time.

(6)

Constitution of a sub-committee for detailed study of various matters pertaining to the Distribution Code and circulation of the findings and recommendations to Review Panel Members and the entities concerned.

(7)

Making arrangements for deliberation of the issues (regarding sub committee findings and recommendations) in the Review Panel meetings the time frame, as provided by these sub-committees.

(8)

Holding of meetings as required but at least one meeting shall be held in every three months.

(9)

Holding of meetings by sub-committees including with any User or with groups of Users to prepare proposals for review panel consideration.

2.7

Review and Revisions. - (1) The Users seeking any amendment to the Distribution Code shall send written requests to the Secretary of DCRP of the Review Panel with a copy to the Commission. If the request is sent to the Commission directly, the same shall be forwarded to the Secretary of Review Panel who shall, in consultation with the concerned entities and such other persons as the Commission may direct, review the Distribution Code provisions. The Secretary of DCRP will circulate the proposed changes/modifications to all the panel members for their written comments within a reasonable time frame or the secretary may call for the Review Panel meeting in consultation with the chairman. Based on this interaction/ discussion, the necessary amendments/revisions may be incorporated in Distribution Code after the approval of Commission.

(2)

The secretary DCRP shall send the following reports to the Commission at the conclusion of each review meeting of the panel:

(a)

Reports on the outcome of such review.

(b)

Any proposed revision to the Distribution Code and its rationale.

(c)

All written representations and objections submitted by the Users at the time of review.

(3)

All revisions to the Distribution Code require the approval of Commission. The secretary DCRP shall publicise revisions to the Distribution Code, after the approval of Commission. The Review Panel may also submit proposals for relaxation in such cases where Users/Distribution Licensees have difficulties in meeting the requirements of the Distribution Code.

(4)

Any change from the previous version shall be clearly marked in the margin. In addition, a revision sheet shall be placed at the front of the revised version, noting the number of every changed sub-section, together with reasons for such change.

(5)

The secretary DCRP shall maintain copies of the Distribution Code incorporating the latest amendments and shall make it available at a reasonable cost to any person requiring it.

(6)

The Commission, may, on the application of the Distribution Licensee or otherwise, call the emergency meeting of the review panel as and when the situation so dictates and make such alterations and amendments in the Distribution Code as it thinks fit.

Chapter 3

Distribution System Planning Code

3.

1. Introduction.

(1)

This chapter specifies the guidelines for planning methodology for the Distribution System and covers following-

(a)

Load Data

(b)

Load Forecast

(c)

Power System Studies and Network Expansion Plan

(d)

System Adequacy and Redundancy

(e)

Security Standards

(f)

Standardisation of sub-station layout

(g)

Standardisation of Nomenclatures and Identification Coding

(h)

Standardisation of design of Distribution Transformers

(i)

BIL and BSL

(j)

Reactive Compensation and Harmonics

(k)

Metering

(l)

Demand Side Management

(m)

Database Management

(n)

Energy Audit

(2)

The Distribution System shall be planned and developed in such a way that the system should be capable of catering the requirement of all categories of Consumers with a safe, reliable, economical and quality supply of electricity, and shall take into consideration the future growth requirements. However, the Consumer shall extend full support to the Distribution Licensee to enable the Distribution Licensee for quality supply of electricity. The Distribution System shall conform to the statutory requirements of all the relevant code, standards and acts in force.

(3)

While planning distribution system, Users and Distribution licensee shall fulfil the requirement stipulated in Regulations issued by CEA under sub-section (2) of Section 177 of the Act and amendments issued from time to time.

3.2

Objectives. - The main objectives of the Distribution System Planning Code are:

(1)

To enable the planning, design and construction of the Distribution System for a safe, reliable and economical operation conforming to Statutory Acts and Rules, which are in force.

(2)

Specify technical conditions to be followed by the respective Distribution Licensees and Users in meeting the standards for an efficient operation of the common electrical interface.

(3)

To prescribe the procedure for the exchange of the system planning data between the Distribution Licensee and the Users for facilitating synchronised planning of the Distribution System at Distribution Licensee and User levels.

(4)

To prescribe the procedure to Distribution Licensee to make Demand Side Management (DSM) an integral part of their day-to-day operations, and undertake planning, designing and implementation of appropriate DSM programmes on a sustainable basis.

(5)

These guidelines of planning cover the individual sub-stations, system planning, analysis and the techno economical aspects in the field of Distribution systems. It applies to all the consumers already connected or awaiting or seeking connection to the distribution system, Distribution Licensees and State Transmission Utility (STU), wherever it is applicable.

(6)

The items mentioned at Regulation 3.1 (1) are discussed in the following paragraphs of this chapter.

3.3

Load Data. - (1) From the metered data collected at each interface point with the transmission system, the Distribution Licensee shall develop load curves for the area fed and also the system load curve for the area of supply in point by applying a suitable diversity factor.

(2)

The Users with Demands of 1 MVA and above shall furnish their load data/ characteristic and other relevant details to the Distribution Licensee as detailed in Annexure 1. The Distribution Licensee shall exercise special care to monitor the actual development of loads in respect of Consumers desiring to avail loads of 1 MVA and above at a single point.

(3)

The Distribution Licensee on his part shall maintain relevant data for designing and selecting the electrical equipment, details of metering and relay for purpose of protection and System data in his Distribution System. The Distribution Licensee shall update the system data regularly and at least once a year.

3.4

Load Forecast. - (1) The Distribution Licensee shall formulate a rolling short-term demand forecast annually for a period of five years in his Area of Supply (to enable the STU in drawing down the annual planning process corresponding to a 5 years forward annual plan for intra state transmission system).

(2)

Energy sales in each tariff class shall be projected in the forecast period after considering the previous financial year as base and projecting the demand for the succeeding 5 years by adopting suitable methodology, such as considering the trend for previous five years and considering the expected economic and social development of various sectors in his Area of Supply in succeeding five years.

(3)

During this process Distribution Licensee shall also review the status of loads materialising as per the previous load forecast. Further these forecasts shall be in line with the plan to be developed at national level by CEA. The Distribution Licensee shall incorporate the variation to the forecast, as and when required.

(4)

The peak load requirements at each Interface Point shall be estimated. However, if the Distribution Licensee receives power at a number of Interface Points in a compact area, which are interconnected in a ring, then the Distribution Licensee shall forward the overall short term Demand forecast at each Interface Point with the variation or tolerance as mutually discussed and agreed upon with the STU.

(5)

Apart from the short term demand forecast for each Interface Point for peak load requirement the Distribution Licensee shall also forward the aggregate energy and peak load demand for Area of Supply on annual basis to the STU, Transmission Licensee (TL) and Commission along with the following details on the basis of which the forecast is made - Data, Methodology and Assumptions.

(6)

The peak load requirement at each Interface Point will essentially ensure that the STU may determine the corrective measures to be taken to maintain the capacity adequacy in the Transmission System up to the Interface Point. This will facilitate the Transmission Licensee to develop the compatible Transmission System.

(7)

The Distribution Licensee shall create a database of loads for each consumer category and for each distribution sub-station and update it annually.

3.5

Power System Studies and Network Expansion Plan. - (1) Based on the projected load, the Distribution Licensee shall carry out the power system studies (load flow analysis) before undertaking major distribution expansion plan on long-term time scale. The distribution licensee shall share its findings of the power system studies with the State Transmission Utility on regular basis for better coordinated power system planning.

(2)

The Distribution Licensee shall employ the software tools for distribution network analysis for:

(a)

Optimum distribution transformer locations;

(b)

Optimum network of sub-transmission system, primary distribution, LT feeders and sub-station location;

(c)

Optimum ratio of the lengths of HT and LT distribution lines;

(d)

Optimum reactive compensation and Harmonic analysis.

3.6

System Adequacy and Redundancy. - (1) The Distribution Licensee while planning Distribution System shall take into consideration the adequacy and redundancy of system capacity and capability to allow for long term load growth based on perspective plan and maintaining supply to consumers in the event of forced or planned outage of lines and transformers. The system shall have built in redundancy so that consumers face no interruption in power supply through alternative circuit arrangements.

(2)

Sub-station design shall allow taking out any transformer for maintenance without affecting supply to any area even during peak hours. More than one transformer with smaller capacity to be employed rather than one transformer of large capacity to meet N-I planning criteria. Alternative circuits shall be planned for important loads. So far as possible, redundancy should be in the system to meet the emergencies and system adequacies shall be taken care of at planning stage of new sub-station(s).

(3)

There shall be at least two numbers of transformers of similar rating in every 33/11 kV Sub-Station.

(4)

In every Sub-Station of capacity 10 MVA and above there shall be a provision for obtaining alternate 33 kV supply to the Sub-Station in case of a failure in the incoming supply.

3.7

Security Standards. - The Distribution System shall be planned and maintained so as to fulfil the following security standards except under Force Majeure conditions beyond the reasonable control of the Distribution Licensee.

(1)

The feeders, either HT or LT, feeding important loads such as hospitals, crematoria, airports, railway stations, and the like shall be planned to have a selective switching system, so that selective switching can be operated to transfer the load on to an alternate healthy feeder. Appropriate safety precautions shall invariably be taken in this regard. In case of failure of the feeder, these switches shall be operated immediately either manually or automatically depending on the importance of the load.

(2)

The rupturing capacity of the switchgear employed in the system shall have at least 25% more capacity than the short circuit level computed even considering the anticipated future development of the system.

(3)

Efforts shall be made for every HT feeder, either primary or secondary, to manually switch over to the immediately available HT feeder of the same voltage class available in the vicinity. Provision shall be made in the design itself for all critical HT feeders to share at least 50% of the loads in the adjacent feeder during emergencies. The same shall be extended to all the HT feeders in a phased manner.

(4)

In case of single contingency; failure of any sub-station equipment controlling any outgoing 11 kV or 33 kV feeders, the load interrupted shall not generally exceed 20% of the total demand on the sub-station. It will not be applicable to the remote inaccessible and snowbound area.

(5)

In order to protect important assets of IT infrastructure, all licensees must adhere to IT Security and Audit Policies issued by the government from time to time. All licensees shall conduct Vulnerability Assessment Audit Exercise and implement Information Security Management System (ISMS) in line with the latest information security standards and must establish Disaster Recovery Plan.

3.8

Standardisation of Nomenclatures and Identification Coding. - The Distribution Licensee shall prepare equipment nomenclatures and identification equipments for uniquely identifying various equipments in distribution system. The nomenclatures scheme shall be consistent with the scheme provided in the UERC (State Grid Code) Regulations, 2016 as amended from time to time for the intra State Transmission System.

3.9

Standardisation of sub-station layouts. - The Distribution Licensee shall develop standard layouts for the sub-stations of 33/11 kV and 11/0.4 kV duly complying with the requirements as specified in the Central Electricity Authority (Technical Standards for Construction of Electrical Plants and Electric Lines) Regulations, 2010.

(1)

11 KV/433 V-3 Phase Distribution Transformer Centers:

(a)

The Distribution transformers up to 160 KVA capacity other than those meant for indoor application shall normally be pole mounted.

(b)

The layout of distribution transformer centres shall generally conform to the relevant REC Construction Standards.

(c)

The distribution transformers shall be located close to the electrical load centre of the load fed by it.

(d)

The distribution transformers above 160 KVA capacity shall be plinth mounted.

(e)

Suitable measures shall be taken sufficiently in advance, to augment the capacity of the feeders and installation of additional transformer centres in the event the specified voltage regulation limits are exceeded.

(2)

All the 33 KV and 11 KV feeders and secondary side of Distribution Transformers shall be provided with electronic energy meters having a memory of 15/30 minutes load survey and billing parameters for 65 days with a provision of RS-232 Port to download data from the electronic energy meters:

Provided that electronic energy meters on all the secondary sides of Distribution Transformers shall be provided in Urban areas within 1 year and in rural areas within 2 years from the date of notification of these regulations.

3.10

Standardisation of design of distribution transformers. - (1) The size/capacity of distribution transformers shall be as per the relevant Bureau of Indian Standards. While the selecting the transformer, due regard should be given to the star ratings issued by the BEE depending upon the filed conditions. As an initial step, the various technical parameters required for the design shall be incorporated in the specifications based on the experience gained regarding the performance among the various designs so far adopted. Later, standard designs of the transformers and their detailed construction drawings shall be evolved based on the performance of these transformers. These shall be adopted for future procurement. This also ensures the interchangeability of components of similar transformers manufactured by different manufacturers.

(2)

Quality Control and Quality Assurance. - A strict quality assurance and quality system management shall be enforced to facilitate the quality of equipment and materials. The Quality Management System to be followed shall be built around a philosophy of "prevention" rather "detection and cure". The various steps involved in the Quality System Management are:

(a)

Inclusion of quality requirement in the contract and selection of good quality vendors/sub-vendors.

(b)

Approval of unambiguous Manufacturing Quality Plan (MQP).

(c)

Finalisation of Field Quality Plan (FQP) ensuring regular, timely and consistent inspection at various stages, viz., raw material, during in-process stage and final inspection and testing prior to dispatch.

(d)

Analyse the equipment failures in association with Engineering and Operation services departments and use feedback for improvement of system.

(e)

Implementation of Quality Systems and Procedures as per ISO-9001 for system of vendor and sub-vendor approvals.

(f)

The list of approved sub-vendors are to be maintained for all the equipment, materials supply and erection works irrespective of the fact whether the Technical Qualifying Requirements are specified in the contract or not. The contractors can choose any sub-vendor from the list of a large number of sub-vendors, with a full transparency. The list is to be continuously revised based on the feedback obtained from the inspection reports, surveillance audits and failure reports, etc.

(3)

Manufacturing Quality Plan

(a)

A standard format is to be developed for the approval of Manufacturing Quality Plan (MQP) which includes the quality requirements at the raw material stage, in-process testing and final inspection and testing requirements as per technical

specifications of the contract and good engineering practices of the industry.

(b)

This document has to be self-sufficient and it should include sample size, acceptance norms, place of testing, requirements of test reports and hold point beyond which the work can progress only after clearance from the utility by standardising the various testing requirements and procedures. It is to be ensured that it is not biased towards any particular manufacturer. These MQP(s) may be approved for a period of three years instead of approval on contract-to-contract basis.

(c)

A good Quality Assurance Plan (QAP) shall be aimed at the following:

(i)

Good quality of raw materials;

(ii)

Quality control during manufacturing and routine tests;

(iii)

Acceptance tests at the time of taking delivery;

(iv)

Inspection and tests on transformers received at stores on random sampling.

3.11

Basic Insulation Level (BIL) and Basic Switching Insulation Level (BSL). - (1) All the equipment in the sub-stations shall be designed to withstand the BIL/BSL values:

Parameter

33kV

11kV

0.415kV

Nominal System Voltage (kV)

33

11

0.42

Highest System Voltage (kV)

36

12

0.45

System Earthing

Solidly earthed system

Solidly earthed system

Solidly earthed system

Lightning Impulse withstand Voltage (KV Peak)

(BIL) 170 75 -

Power Frequency withstand Voltage (KV rms) in

dry condition (BSL)

75

28

3

Note. - The values of BIL/BSL given in the table above are for altitude up to 1000m above MSL. However, for altitudes higher than 1000 m above MSL, BIL/BSL requirements shall be kept higher as per relevant standards and practices.

(2)

The Basic Insulation Level (BIL)/Basic Switching Impulse Insulation Level (BSL) of the equipment to be installed in the distribution system shall be adequate to withstand the lightning/switching surges respectively. Lightning Arresters shall be provided for all the Transformers (33/11 kV and distribution Transformers 11/0.4 kV) and 33 kV and 11 kV lines. The lightning protection system to other equipment in the sub-station by shield wires or lightning masts shall be provided.

(3)

Besides the voltage levels mentioned above the Commission may specify other voltage levels to be a part of the Distribution System under extraordinary conditions.

3.12

Reactive compensation and Harmonics. - (1) Shunt capacitors un-switched/ switched type, shall be installed at the appropriate places in the Distribution System for minimising reactive power drawl from the grid, power factor improvement, maintaining satisfactory voltage profile and reduction of sub-transmission and Distribution losses. The size and location of the capacitor installations shall be determined using appropriate computer software, with reliable field data. Suitable precautionary measures, such as automatic switching, etc., shall be adopted to avoid over voltages during light load periods.

(2)

Optimisation studies of shunt compensation shall be conducted by the Distribution Licensee to determine the most appropriate sizes and locations for shunt capacitor installations.

(3)

The Harmonic measurements shall be carried out periodically by the Distribution Licensee and action taken to reduce the harmonics accordingly.

(4)

The total harmonic distortion for voltage and connection point shall not exceed 5% with no individual harmonic higher than 3%.

(5)

The total harmonic distortion for current drawn from the transmission system at the connection point shall not exceed 8%.

(6)

The respective User responsible for generating harmonics adversely affecting the distribution system shall be responsible for appropriate correction.

(7)

The measurement of harmonics and analysis should generally be in accordance with IEEE 519 guidelines or the regulations specified by the CEA.

3.13

Metering. - (1) All interface meters, consumer meters and energy accounting and audit meters shall be installed and operated in conformance with the Central Electricity Authority (Installation and Operation of Meters) Regulations, 2006/Metering Communication and Data Acquisition Requirements (MCDAR) for Intra-State Power System of Uttarakhand as amended from time to time.

(2)

The renewable energy generators connected to the distribution network are required to provide and maintain meters and communication system for ensuring transfer of data to SLDC in accordance with the CEA Regulations and directions issued by CEA/Commission from time to time.

(3)

The metering for 230 V single-phase supply shall be provided on a board or within a suitable box located in such a place protected from sun and rain and shall be in a convenient position for taking readings. The terminals of the meter shall be made tamper-proof and sealed. For 400 Volts three phase supply, the meters and associated metering equipment including connections shall be enclosed in a suitable tamper-proof box. The tamper-proof box shall be of sufficient strength and design with locking and sealing devices and shall have adequate provision for heat dissipation with the required electrical clearances. The design shall permit readings to be taken without access to its connections.

(4)

For HT Consumers the meters, maximum demand indicators shall be housed in a separate metering compartment and secondary apparatus such as instrument transformers and connections required shall be housed in a separate compartment, which shall be locked/sealed to prevent tampering.

(5)

The HT metering cubicle shall be suitable for cable entry on both sides or at leaston one side. No fuses are permitted in the secondary circuits of the instrument transformers. The metering cubicle shall be painted with suitable epoxy paint for installation in snowbound areas and other areas experiencing heavy rainfall. The instrument transformers shall be of

fixed ratio and shall not have any taps. The primary current rating of the current transformers shall match with the normal full load current and the saturation point of the core shall be higher than the maximum current that may occur due to simultaneous full load operation of all the connected equipment and machinery.

(6)

For HT and EHT Consumers, the secondary terminals of the instrument transformers shall be locked and sealed and the secondary wires brought out in a suitable GI conduit pipe up to the metering panel. There shall be no joints in the conduit pipes. The meters shall be as close to the instrument transformers as possible and in no case shall exceed ten (10) meters. The metering panel shall be housed in a weatherproof and tamperproof box and sealed in conformance with the Central Electricity Authority (Installation and Operation of Meters) Regulations, 2006/Metering Communication and Data Acquisition Requirements (MCDAR) for Intra-State Power System of Uttarakhand as amended from time to time.

(7)

Provision for remote/automatic meter reading of LT, HT and EHT installations shall be provided as required by the system operator or wherever considered necessary by the Distribution Licensee.

(8)

Prepaid meters installed at the consumer premises should be capable of updating of prepaid softwares/logics and in no way should cause any hindrance in implementation of annual tariff revisions.

3.14

Demand Side Management (DSM). - (1) The DSM objectives, for advancement and implementation of cost effective DSM initiatives in the State are sought to be achieved through, but not limited, to the following-

(a)

To make DSM an integral part of the day to day operations of the Distribution Licensee with the objective of improving end use efficiency of electricity, environmental conservation and cost reduction.

(b)

To promote and implement Load Shifting, Power Shortage Mitigation, seasonal peak reduction, cost effective energy savings, lowering the cost of electricity, reduction in emissions of green house gases and pricing initiative i.e. Time of Use/ Seasonal, etc.

(c)

To complement supply side strategies in order to avoid, reduce or postpone investments in generation, transmission and distribution infrastructure.

(d)

To reduce transmission and distribution losses through implementation of end use energy efficiency programmes in areas reporting high AT&C losses.

(e)

To reduce the environmental damage by reducing the emission of green house gases.

(f)

To encourage consumers to revise their electricity consumption pattern both with respect to timing and level of electricity demand for efficient use of energy.

(g)

To promote the use, and create awareness about Solar Pumps, Solar Lights, LED lamps, etc.

(2)

DSM Plan and Programme preparation, submission and approval by the Commission:

(a)

Distribution Licensee shall formulate a perspective DSM Plan and submit the same to the Commission for in-principle approval for the specified period and as per the time lines that may be specified by the Commission from time to time;

(b)

The perspective DSM plan shall include:

(i)

A section outlining the goals sought to be achieved in the specified period.

(ii)

Annual DSM Plan for each year of the perspective plan period containing description of DSM programmes proposed to

be undertaken and its prioritisation along with programme wise details like-

? indicative cost & its cost effectiveness pursuant to these Regulations and any guidelines issued by the Commission from time to time;

? indicative baseline data and expected reduction in tariff and/or gain in energy efficiency;

? consumer segments and estimated level of participation;

? implementation strategy and schedule including process/mechanism for execution e.g. Energy Service Companies, Bidding for DSM Projects, DSM Resource Acquisition, etc.;

? expected payback period and mechanism for recovery of cost and performance incentives.

(iii)

Mechanism for Monitoring, evaluation and reporting of perspective plan and annual plan/programmes;

(iv)

Programme/Plan for Training/Seminars/Workshops to increase consumer awareness.

3.15

Database Management. - (1) The availability of accurate and reliable data is essential for planning and development of the Distribution System on long-term basis. Data management system facilitates storage, retrieval and updating of data for complying with the requirements of the Distribution Code and for other purposes like power system studies.

(2)

The embedded generators connected with distribution system or seeking new connections shall furnish planning data in format specified at Annexure 2. The large consumers connected to or seeking connection to HT or F.11T and having connected load of 1 MVA or above shall furnish planning data in prescribed manner as at Annexure 1 for long term planning by Distribution Licensee. Distribution Licensee shall supply system data to the Users, embedded generators and large consumers, wherever required for their planning purpose as per format at Annexure 3.

(3)

A well maintained Data Management System would facilitate exchange of data between Users and Distribution Licensee required for long-term planning and distribution operation in an accurate and reliable manner. This will also help in Users, Large Consumers, open access customers and embedded generators to have access to data, which they may require for their planning purpose.

(4)

Consumer Indexing:

(a)

The last mile of a distribution network is the pole or support in overhead system or the service pillar/feeder pillar box in underground cable distribution system. Hence, the consumer indexing has to be done with respect to the last mile of the distribution network. The integration of consumer indexing/information with distribution transformer wise in a distribution network system is key to providing good consumer services.

(b)

The integration of consumer index with the distribution network will help the distribution licensee to operate the distribution network better in the manner to simulate the network to estimate the voltage profile across the network and identify low voltage pockets without actually visiting consumer installation and measuring voltages. This will also facilitate distribution licensee in conducting energy audit by estimating correctly the energy sales on the 11 kV Feeder/distribution transformer and account for energy supplied to the 11 kV Feeder/distribution transformer.

3.16

Energy Audit. - (1) The Distribution Licensee shall create responsibility centres for energy audit. Distribution sub-division in charge and division in charge shall be made as responsibility centres and countable for the energy input and sales in their respective areas. They shall also compute month/year wise distribution losses and prepare energy balance sheets of their respective areas.

(2)

The Distribution Licensee shall establish and maintain a system for segregation of technical and commercial losses through energy audits. Interface meters capable of data retaining capacity of at least 65 days shall be installed for all the incoming/outgoing feeders for each such unit.

(3)

The energy audit for total system shall be carried out by compiling the data and analysis carried out in each

responsibility centre. The energy received from each sub-station shall be measured at the 11 kV/33kV terminal switchgear of all the outgoing feeders installed with appropriate energy meters such that the energy supplied to the each feeder is accurately available. It shall be compared with the corresponding figures of monthly energy sales and the distribution loss for each feeder shall be worked out. In case the Distribution Licensee has adopted ring main system at 11 kV and 33 kV and there is difficulty in determining the distribution losses for each feeder, then the Distribution Licensee shall work out distribution losses for the overall Area of Supply.

(4)

An action plan for reduction of the losses with adequate investments and suitable improvements in governance should be drawn up and shall be submitted to the Commission annually along with Annual Revenue Requirement Filing.

Chapter 4

Connectivity Conditions

4.

1. Introduction.

- The Connectivity Conditions specify the minimum technical and design criteria, which shall be complied by any user connected to, or seeking connection to the Distribution System. The Distribution Licensee shall ensure compliance of the above criteria by any user as a pre-requisite for the establishment of an agreed connection. In addition to the above, the connectivity conditions should fulfil the requirement stipulated in Section 50, Section 53 and regulations issued under sub-section (2) of Section 177 of the Act.

4.2

Objectives. - (1) The objectives of the Connectivity Conditions are to specify that:

(a)

The basic rules for connections are complied by all user. This will help to treat all user in a non-discriminatory manner.

(b)

Any new or modified connection, when established, shall not suffer unacceptable effects due to its connection to the Distribution System nor produce unacceptable effects on the system or any other connected user.

(c)

The ownership and responsibility for all the equipments, in case of interface/ connections with all Users at HT and EHT, shall be clearly specified in a Site Responsibility Schedule as per Format specified in Annexure 4 for every site, where a connection is made.

4.3

Interface Point. - (1) Connection to Transmission System shall be governed by the UERC (State Grid Code) Regulations, 2016 as amended from time to time.

(2)

Connection of Generators and Captive generators including DG sets to Distribution Systems may be provided at bus bars of the generating station. All generating units shall inject the output into the bus bars via the synchronising breaker. The isolator between the synchronising breaker and the bus bars would be the boundary between the generator and the Distribution Licensee. The current transformers of the tariff metering shall be connected near the synchronising breaker. The voltage transformers (including the standby set) of the tariff metering shall be connected to the bus bars.

(3)

EHT/HT Consumers: The supply voltage may be 220kV/132kV/66kV/33kV/11kV or voltage as agreed by the Distribution Licensee. The interface point for the HT/EHT consumer shall be the Distribution Licensee's cut off point/isolator in the consumer's switchyard at consumer's end. However, in case EHT/HT consumer is being fed from a dedicated feeder emanating from Transmission/Distribution Licensee's sub-station, the interface point shall be the line isolator at the said sub-station of the Transmission/Distribution Licensee as the case may be.

(4)

Low voltage consumers: The incoming terminal of the cut out/circuit breaker installed by the consumer is the boundary of low voltage Consumers. The tariff metering shall be provided before a fuse unit/circuit breaker of the Consumer. The metering equipment shall be provided at the entry point of Consumer Premises in a safe location, preferably at the entry of the boundary of the premises or in a common passage on ground floor or nearby safe location outside the premises for easy access for the purpose of meter reading, maintenance, repairs, inspection, etc. The metering equipment shall be provided inside a box sealed by the Distribution Licensee and the User/Consumer shall not disturb the seal of the

metering equipment and shall take reasonable care for upkeeping protecting the meter and equipment.

4.4

Operational Labelling. - (1) The Distribution Licensee and the Users shall be responsible for the provision and maintenance of clear, unambiguous signs and labels indicating the numbering and/or name of the equipment/apparatus and circuit at the sub-stations and connection sites.

(2)

The equipment installed shall conform its relevant I.S. specification and the ratings and salient specification shall be maintained on the equipment's nameplate. No electrical equipment shall be used without its manufacturers nameplate permanently affixed to it.

4.5

System Performance. - (1) The design and construction of all the equipments connected to the Distribution System shall satisfy the relevant Indian Standard Specifications to the maximum extent possible.

(2)

Installation of all electrical equipment shall comply with rules and code of practice in force.

(3)

For every new connection sought, the Distribution Licensee shall specify the Connection Point/Interface Point and the supply voltage, along with the metering and protection requirements as specified in the code.

(4)

The operation of the Distribution System shall be in accordance with the "Operation Code". The User shall however be subject to the distribution discipline prescribed by the SLDC/ ALDC.

(5)

The insulation co-ordination of the Users' equipment shall conform to the Regulation 3.1.1 of these regulations.

4.6

Procedure for Applications for connections to the System. - Any User seeking use of Distribution System is required to submit application for connection to the Distribution Licensee as per the procedures and formats laid down by the Commission in the relevant regulations for release of new LT/HT/EHT connections.

4.7

Connection Agreement. - A Connection Agreement between User and the Distribution Licensee shall be executed both for sale and/or purchase including Independent Power Producer (IPP). For LT consumers, an undertaking is required to be furnished as per UERC (Release of new LT Connections, Enhancement and reduction of Loads) Regulations, 2013 as amended from time to time.

Chapter 5

Operation Code

5.

1. Introduction.

- This chapter contains the procedures and practices to be followed for safe and efficient operation of the Distribution System by the Distribution Licensee and Users. While operating distribution system, Users and Distribution licensee shall fulfil the requirement as stipulated in regulations issued by CEA under sub-section (2) of Section 177 of the Act and amendments issued from time to time.

The following aspects of operation are covered in this section:

(1)

Demand Estimation

(2)

Outage Planning

(3)

Contingency Planning

(4)

Demand Management and Load Shedding

(5)

Interface with Generating Plant including CPPs

(6)

Monitoring and Control of voltage, and power factor

(7)

Requirements with respect to Harmonics, Direct Current (DC) injection and Flicker

(8)

Safety Co-ordination

(9)

Communication

(10)

Maintenance and testing

(11)

Tools and spares

(12)

Training

5.2

Objective. - The objective of this chapter is to define the operating procedures and practices to be followed by the Users and Distribution Licensee for ensuring Distribution System safety, stability, efficiency and security.

5.3

Demand Estimation. - (1) The Distribution Licensee shall estimate its requirement (demand and energy) for 15 minutes blocks at Transmission - Distribution Interface points on day-ahead/week-ahead/month-ahead basis for his Area of Supply based on relevant load curves subject to modifications depending upon the communication received from any specific User or caused by any contingency. This may be given to SLDC as per its requirements.

(2)

For this purpose, the concerned major Users identified by the Distribution Licensee shall furnish the data pertaining to their demands to the Distribution Licensee.

5.4

Outage Planning. - (1) The Distribution Licensee shall furnish his proposed outage programs to the Transmission Licensee on a month ahead basis. The outage program shall contain identification of lines and equipment of the Distribution System proposed by Distribution Licensee.

(2)

The outage plan proposed by the Distribution Licensee shall come into effect only after the Transmission Licensee releases the finally agreed transmission outage plan.

(3)

However, at the time the line or equipment of the Distribution Licensee is taken out of service, the Distribution Licensee shall intimate the Transmission Licensee that the Transmission Licensee may also carry out its maintenance works pertaining to the line/ equipment under maintenance, if required, so that the maintenance works could be carried out under minimum shutdown period.

(4)

In case of lines and equipment of 66 kV and above the specific concurrence of SLDC shall be obtained in addition to the above.

(5)

The above procedure shall not apply under the following circumstances:

(a)

Emergency situations' to save plant and machinery;

(b)

Unforeseen emergency situations requiring isolation of lines or equipment to save human life;

(c)

Where disconnection is to be effected on any User installation due to breach of Agreement. In this case the SLDC shall be informed wherever the load to the extent of 1 MVA or more is affected.

(6)

Planned outages of Power System for the duration as specified in UERC (Standards of Performance) Regulations, 2007 as amended from time to time for maintenance purposes shall be intimated to the public through media including

publishing in at least two largely circulated Newspapers In Uttarakhand (one in Hindi and one in English) of that area, two days in advance.

5.5

Contingency Planning and Crisis Management. - (1) A contingency situation may arise in the event of a total or partial blackout in the Transmission System. A contingency may also arise on a part of the Distribution System due to local breakdowns in the Distribution System itself. It may also arise due to a breakdown in the Apparatus of the Transmission Licensee at the point of interconnection.

(2)

Contingency and crises management procedure shall be documented unambiguously to achieve the restoration of the total system and associated demand, and re-synchronisation of parts of the total system, which have become out of synchronism with each other, at the shortest possible time.

(3)

Transmission System Failure:

(a)

In case of a total blackout at any point in the Area of Supply of Distribution Licensee, the Distribution Licensee shall follow the black start procedures framed by the Transmission Licensee.

(b)

The Distribution Licensee shall sectionalise the Distribution System into discrete blocks of demand. The Distribution Licensee shall advise and co-operate with the SLDC for the amount of MW load likely to be picked up on switching each demand block.

(c)

The Distribution Licensee shall prepare a schedule of essential and non-essential loads in order of priority at each connection to be picked up during the restoration process and shall furnish the same to SLDC.

(d)

The Distribution Licensee shall maintain direct communication links with the SLDC and shall ensure and maintain the load generation balance under the direction of the SLDC.

(e)

The Distribution Licensee shall furnish the names and designations of the person(s) with their contact details viz. telephone/mobile numbers and e-mail id and stations, authorised to deal with contingency operations, to the SLDC.

(4)

Failure of the Apparatus of the Transmission Licensee:

(a)

The Distribution Licensee shall immediately contact the authorised person at the sub-station of the Transmission Licensee, and assess the probable period of restoration and the probable restriction of load drawl from the affected sub-station.

(b)

The Distribution Licensee shall effect the demand management plan accordingly.

(5)

Distribution System Failure:

(a)

interruptions to power supply in any part of the Distribution System lasting for the period as specified in UERC (Standards of Performance) Regulations, 2007 as amended from time to time for Distribution Licensee due to breakdown in any part of the Distribution System may be termed as a Distribution System Failure.

(b)

The Distribution Licensee shall coordinate with SLDC for restoration process, which shall be as per UERC (State Grid Code) Regulations, 2016 as amended time to time.

(c)

The Distribution Licensee shall designate a Nodal officer to coordinate with SLDC for distribution system restoration.

5.6

Demand management and load shedding. - (1) Temporary Load Shedding may be resorted to, for maintaining the grid frequency as instructed by the SLDC. Temporary Load Shedding may also be necessitated due to loss of any circuit or

equipment or any other operational contingency. In case of automatic load shedding through Under Frequency Relays, the circuits and the amount of load to be interrupted with corresponding relay settings shall be co-ordinated with the SLDC and persons in charge of the sub-stations of the Distribution Licensee as necessary.

(2)

In the event of sustained shortages, Distribution Licensee shall submit a detailed program for planned load shedding indicating areas and timing of proposed load shedding for approval of the Commission. On receiving Commission's approval, Distribution Licensee shall publish the approved program in atleast two local newspapers. Distribution Licensee shall seek fresh approval from Commission for any deviation from approved load shedding program.

(3)

If the duration of unplanned load shedding to any part of the Distribution System exceeds 2 hours, the affected Consumers on independent circuits, emanating from primary sub-stations may be suitably intimated. The essential services such as public hospital, public water works, sewage works, etc., shall be intimated over the telephone/mobile/other electronic communication medium wherever possible.

(4)

Dedicated feeders for supply of power to agriculture consumers shall be constructed so as to ensure 8 to 10 hours supply on such feeders.

(5)

33 KV ring mains shall be provided in towns.

5.7

Interface with Generating Units including Captive Power Plant (CPP). - (1) If the Distribution Licensee has an interface with any generating unit including CPP and an Agreement for this purpose exists, the Distribution Licensee and the concerned owner of the generating unit shall abide by the following provisions in addition to the provisions contained in this Code as applicable to all the Users:

(a)

The owner shall provide suitable protection at the interface to protect his system from any damage due to normal and abnormal conditions in the Distribution System.

(b)

If the generator is an induction generator, the owner shall take adequate precautions to limit the system disturbances, when the induction generator is synchronised in consent with the Distribution Licensee. Generating Company having induction generators shall install adequate capacitors to compensate the reactive power drawl. Also whenever the power factor is found very low during starting period and causes voltage dip in the Distribution Licensee's system the Distribution Licensee may advise the owner to install capacitors and the generating company shall comply. Failure to comply entails penalty and/or disconnection from the system as per provision of Act/Rules/Regulations.

(2)

The owner shall comply with the provisions of the UERC (State Grid Code) Regulations, 2016 as amended from time to time.

5.8

Monitoring and Control of Voltage and Power Factor. - (1) The Distribution Licensee shall monitor the voltage and power factors in the Distribution System at system input points at peak and off-peak hours and take reasonable measures for improvement of the same in coordination with the Users with demand of 1 MVA and above, and the Transmission Licensee.

(2)

The Distribution Licensee shall take power factor improvement measures at strategic points in the Distribution System by carrying out system studies and installing the required reactive power compensation equipment.

(3)

Users having loads with low Power Factor shall install capacitors of appropriate rating as per Annexure 5. The consumers using power for welding purposes shall install the equipment as specified by Distribution Licensee from time to time, to prevent frequent voltage fluctuations.

(4)

The Distribution Licensee and Users shall abide by the instructions issued by the SLDC from time to time on load management for maintaining the frequency of supply within the specified limits.

5.9

Requirements with respect to Harmonics, Direct Current (DC) injection and Flicker. - (1) Voltage and current harmonics: In any installations the voltage and current Total Harmonic Distortion (THD) shall not exceed the limit as specified in the Central Electricity Authority (Technical Standards for connectivity to the Grid) Regulations, 2007, as amended from time to time, applicable to the Distribution Systems. The Consumer will take appropriate measures to restrict the THD to the stipulated limit.

(2)

The Generating station(s) shall not inject DC current greater than 0.5% of full load rated output at the interconnection point.

(3)

The Generating Stations shall not introduce Flickers beyond the limits specified in IEC61000

(4)

Measurement of Harmonic Contents, DC injection and Flicker shall be done at least once in a year in the presence of parties concerned.

5.10

Safety co-ordination. - (1) The Distribution Licensee and the Users (comprising Generating Companies, Transmission Licensee and Consumers having load above 1MVA or dedicated lines) and any other Distribution Licensee having common electrical interface with the Licensee shall designate suitable persons to be responsible for safety co-ordination. These persons shall be referred to as Safety and Control Persons. Their designations and telephone/mobile numbers shall be exchanged between all the concerned persons. Any change in the list shall be notified promptly to all the concerned.

(2)

The Distribution Licensee and Users shall prepare safety manuals incorporating all the safety precautions to be taken for each component of the Distribution System based on the Distribution System under Safety Code specified at Chapter 7 of these Regulations. All the safety rules and precautions shall be observed when work is to be carried out on any line or apparatus, switchgear or circuits in any part of the Distribution System or in any part of the User System. The safety manuals thus prepared shall be issued to all the Safety and Control persons and such Users for compliance.

(3)

There shall be co-ordination between persons of the Distribution Licensee and the Users, between persons of two Distribution Licensees having electrical interfaces, for carrying out the work on any apparatus or lines, etc., belonging to either party at the point of interconnection.

(4)

The provisions of the UERC (State Grid Code) Regulations, 2016 as amended time to time, shall be followed at Connection Points/Interface Points in co-ordination with the Transmission Licensee.

(5)

The disconnecting device(s) at each electrical interface, which shall be capable of effectively disconnecting the system of the Distribution Licensee and the other Users, and the grounding devices of the respective systems at the control boundary shall be identified and marked by the Distribution Licensee and the respective Users. These shall be maintained in good condition at all times. To prevent inadvertent switching operations by unauthorised persons, such disconnecting devices shall be provided with interlocks.

(6)

Wherever any Consumer has installed an emergency power supply system, either an electronic system with storage batteries or with generators, the arrangement shall be such that the same cannot be operated without clearly isolating the system from the supply mains. The responsibility of making the required arrangement for isolation from supply mains shall be of the User and this shall be part of the electrical layout submitted to Electrical Inspector for his approval. A copy of the approved layout shall be provided to Distribution Licensee thereafter.

The possibility of a feedback from these devices to the Distribution System from any of the conductors, including the neutral conductor shall be clearly ruled out:

Provided that the grid synchronised generation shall be exempted from above provisions.

(7)

The appropriate Control Person at the electrical interface shall issue written permission to his counterpart for carrying

out the work on any apparatus, switchgear or lines beyond the electrical interface. Such permissions shall be termed as Permit to Work (PTW). The prescribed format for PTW is annexed as Annexure 7 of these regulations which shall be used by all concerned.

(8)

All maintenance work shall be duly authorised by the designated officer. The system of PTW shall be observed for carrying out any maintenance work. The line should not be energised back without return of PTW after completion of maintenance work.

(9)

The Distribution Licensee in consultation with the concerned User shall frame checklist of operations to be carried out and the procedures for safety coordination for each electrical interface, before issue and return of PTW. Such procedures and checklists shall be issued to all the concerned by the Distribution Licensee for implementation.

5.11

Operational Communication. - (1) Reliable communication such as telephones, mobiles and e-mails, etc., links shall be established for exchange of data, information and operating instructions between SLDC and the Distribution licensee, embedded generators, users and large consumers with a demand of more than 1 MVA.

(2)

The Distribution Licensee and the Users connected to its Distribution System shall designate officers and agree on communication channels for the exchange of information. Communication shall, as much as possible, be direct between the User and the operator of the Distribution System to which that User is connected.

(3)

List of telephone/mobile numbers, call signs and e-mail IDs shall be exchanged by the Distribution Licensee and concerned Users to enable control activities to be efficiently coordinated.

5.12

Mobile Breakdown Vans. - The Distribution Licensee shall provide Mobile Breakdown Vans in towns and cities for attending line and transformer faults and consumers' complaints without any delay. The Mobile Breakdown Vans will be equipped with all necessary Tools such as cable jointing kits and consumable at all times on duty. The Breakdown Vans shall be fitted with wireless phone, telescoping ladder. All spares necessary for maintenance work shall be provided in such breakdown van and inventory of spares shall be replenished from time to time.

5.13

Reserves and standbys. - (1) The Distribution Licensee shall maintain adequate reserves and standby emergency equipment for attending forced outage conditions of lines and transformers in each distribution division. These include oil filtration sets, cable jointing and maintenance kits, mobile cranes, chain pulley, lifter, etc.

(2)

The Distribution Licensee at all time must have adequate stock of spare transformers, isolators, circuit breakers, meters. Instrument Transformers, insulators, hardware, cable and cable boxes, etc., for undertaking emergency works in each distribution division.

(3)

The Distribution Licensee shall have maintenance and live wire maintenance Teams available at important locations which can be called and deployed on maintenance work of emergent nature.

5.14

Construction Practices. - (1) All electric supply lines and apparatus shall be of sufficient ratings for power, insulation and estimated fault current and of sufficient mechanical strength, for the duty which may be required to perform under the environmental conditions of installation, and shall be constructed, installed, protected, worked and maintained in such a manner as to ensure safety of human beings, animals and property.

(2)

The relevant code of practice of the Bureau of Indian Standards including National Electrical Code, if any may be followed. The material and apparatus used shall conform to the relevant specification of Bureau of Indian Standards where such specifications have been already laid.

(3)

The Distribution Licensee shall prepare and observe the Construction and Maintenance Manuals for various equipment/works like 33 kV Lines, 11 kV Lines, LT Lines, 33 kV substation and 11 kV sub-stations. The Construction

and Maintenance Manual shall be prepared taking into consideration the following:

- (a) Technical Standards for construction of electrical plants, electric lines and connectivity to the grid specified by the Central Electricity Authority (Technical Standards of Construction of Electrical Plants and Electric Lines) Regulations, 2010 as amended from time to time;
- (b) Safety requirements for construction, operation and maintenance of electrical plants and electric lines specified by the Central Electricity Authority (Measures relating to Safety and Electric Supply) Regulations, 2010 as amended from time to time;
- (c) REC Construction Standards and Standard design layouts;
- (d) CBIP Publications on Code of Practices;
- (e) Code of Practices issued by Bureau of Indian Standards for various equipment and maintenance practices; and
- (f) Instruction Manuals for installation, operation and maintenance issued by standard equipments manufacturer concerned.

(4) The standard tables for conductor size, fuse size, wire gauge, electrical clearance, ground wire size, insulation resistance and earth resistivity, etc., shall be included in the Construction and Maintenance manual. Distribution Licensee shall ensure that its construction and maintenance staff strictly observe the norms given in the Manuals.

5.15

Preventive Maintenance Schedules. - (1) The Distribution Licensee shall prepare a Preventive Maintenance Schedule for various line and sub-station equipment installed in distribution system. The Preventive Maintenance Schedule shall include the following important equipment:

- (a) Power Transformers and Distribution Transformers installed outdoor/indoor;
- (b) 11 kV and 33 kV Circuit Breakers and associated equipments;
- (c) 11 kV and 33 kV Overhead lines including Gang Operated (GO) Switches and Drop Out Fuses;
- (d) 11 kV and 33 kV Cable and Cable Boxes;
- (e) LT Lines and circuit breakers; and
- (f) Equipments related to Service Connection.

(2) The Preventive Maintenance Schedule shall have sections covering the following;

- (a) Recommended Schedule for inspection;
- (b) Recommended Schedule for preventive maintenance ; and
- (c) Recommended Schedule for overhaul.

(3) The inspection schedule and preventive maintenance schedule shall have daily, weekly, monthly quarterly and annual periodic activity to be carried out for various equipments.

(4) In addition to the preventive maintenance, the Distribution Licensee shall also be required to conduct Technical Audit of

atleast one 33/11 kV Sis in each Electricity Distribution Circle (EDC) in a Financial Year. The Technical Audit shall be conducted through competent Independent Agency(ies) appointed by the Distribution Licensee from time to time for verifying proper functioning of the Distribution Protection System and adequacy check of various elements/assets installed at 33/11 kV Sis and its outgoing feeders upto the secondary side of Distribution Transformers (DTs) as per Schedule given below:

of Activities for Technical Audit

Description of Activities

Financial yearwise schedule

Responsibility

Preparation of Terms of Reference (ToR) for

conducting Annual Technical Audit and Publication of Notice

Inviting Tender for conducting Technical Audit of the identified

33/11 kV sub-stations under the Distribution Code Regulations

By 30th day of April

Distribution Licensee

Selection of Technical Auditor (s) and issuance

of Work Order(s)

By 30th day of June

Distribution Licensee

Submission of Report(s) including Action Points

by the Technical Auditor(s) to Distribution Licensee.

By 30th day of September

Technical Auditor(s)

Action Taken report(s) of Distribution Licensee

on the recommendations made by the Technical Auditor(s) to be

submitted before the Commission every year.

By 31st day of March

Distribution Licensee

5.

16.

Maintenance Record. - (1) The Distribution Licensee shall maintain records of periodic inspections carried out in the standard formats prescribed in Preventive Maintenance Schedule. Records shall be maintained in respect of following amongst others in respective sub-station/sub-division/division office:

(a)

Power Transformers and Distribution Transformers installed indoor/outdoor

(b)

11 kV and 33 kV Circuit Breakers;

(c)

33 kV and 11 kV Lines

(2)

Regular testing of all the equipments such as transformers, switchgear, protective relays, etc., should be carried out as recommended by the manufacturer and the relevant code of practice issued by the Bureau of Indian Standards and CBIP. These shall be carried out at the prescribed intervals and the test results shall be recorded in the maintenance registers maintained at respective sub-stations. Wherever the test results indicate a decline in the insulation resistance and/or deterioration of the equipment, preventive maintenance shall be earned out to ensure serviceability, safety and efficiency.

(3)

The consumers shall maintain their apparatus and power lines at all times conforming to Central Electricity Authority (Measures relating to Safety and Electricity Supply), 2010 as amended from time to time and these shall be suitable for connection to distribution system in a safe and reliable manner.

5.17

Environmental Issues. - (1) The Distribution Licensee shall take due regards of environment regulatory guidelines in planning, design, construction and operation of distribution system. Environmental impact assessment shall be carried out for all major distribution projects like construction of sub-stations in green and reserved area. The required clearances and no-objection shall be obtained from State Pollution Control Boards wherever prescribed.

(2)

The Distribution Licensee shall ensure that the environmental concerns would be suitably addressed through appropriate advance action by way of comprehensive Environmental Impact Assessment and implementation of Environment Action Plan (EAP).

5.18

Energy Conservation. - (1) The Distribution Licensee in order to minimise the overall energy demand shall accord high priority to energy conservation and demand side management (DSM). The Distribution Licensee shall ensure compliance of the Energy Conservation Act, 2001 and adhere to the guidelines of Bureau of Energy Efficiency issued from time to time in this regards.

(2)

Distribution Licensee shall ensure that the periodic energy audits wherever have been made compulsory for power intensive industries, under the Energy Conservation Act, being complied by its consumers. Other industrial consumers may also be encouraged to adopt energy audits and energy conservation measures. Energy conservation measures viz. installation of roof top solar power plants based on net metering, rooftop solar water heater and solar lighting, etc., shall be adopted in all Government buildings for which saving potential has been estimated to be about 30% of the average energy consumption.

(3)

The Distribution Licensee in the agriculture sector shall promote the pump sets and the water delivery system engineered for high efficiency and preferably of appropriate star labelling. In the industrial sector, The Distribution Licensee shall take action for promoting energy efficient technologies as energy conservation measures. Motors and drive system are the major source of high consumption in Agricultural and Industrial Sector. The Distribution Licensee shall advise the consumers to use high efficiency motors in Agricultural and Industrial Sector. Distribution Licensee shall take effective steps such that energy efficient lighting technologies should also be adopted in industries, commercial and domestic establishments.

(4)

Distribution Licensee shall endeavour that the requirements for capacity additions is reduced to the extent possible by reducing the difference between electrical power demand during peak periods and off-peak periods through suitable load management techniques such as differential tariff structure for peak and off peak supply and metering arrangements (Time of Day metering) to achieve an efficient load management.

5.19

Tools and Spares. - (1) The Distribution Licensee shall ensure availability of proper tools and tackles, equipments viz. megger, tong-tester, earth tester, etc., at all work places for carrying out the maintenance. The tools and tackles shall be checked from time to time and their serviceability shall be ensured.

(2)

The Distribution Licensee shall maintain an inventory of spares required for maintenance and replacement purposes at suitable locations according to a clear policy to be laid down by the Distribution Licensee.

5.20

Human Resource Development and Training. - The Distribution Licensee shall impart necessary training to its officers/staff in distribution system operation and maintenance practices so as to implement the provisions of these Regulations. The Distribution Licensee shall make appropriate arrangements for imparting training to workmen and supervisory staff, incorporating up-to-date techniques and safety measures of distribution system design, construction and maintenance. The Distribution Licensee shall prepare an annual training calendar for its officers and staff and shall submit the same to the Commission latest by 15th May of every year.

5.21

Geographical Information System (GIS)/Global Positioning System (GPS) based information system. - (1) The Distribution Licensee shall, in stages, deploy GIS/GPS based Geographical Facilities Information System for planning operation and maintenance of distribution system. The GIS shall be utilised for mapping the all important elements of

distribution system which includes lines, transformers, sub-stations, generating stations, all unit locations and shall eventually covers all consumers. The GIS shall be linked to active Relational Database Management System (RDBMS) and GPS shall be utilised for time synchronisation.

(2)

The digital maps of distribution network shall be developed for each of the following preferably by conducting GPS survey which is easier, fast, accurate and economical:

(a)

33 kV network of complete distribution system indicating distance, type and size of conductor/size of Under Ground (UG) cable with single core or 3 core for lines and sub-station particulars with Single Line Diagram (SLD).

(b)

The feeder-wise 11 kV lines/cables indicating distance, type and size of conductor/ UG cable, location and capacity of distribution transformers.

(c)

The distribution transformer-wise LT line/cables with number of consumers and connected load of each of LT support/LT feeder pillar Box.

Chapter 6

Distribution Protection Requirement

6.

1. Introduction.

- In order to safeguard Distribution System and prevent faults travelling into the Transmission System, it is essential that certain minimum standards for protection shall be specified for the Distribution Licensee and Users connected to Distribution System. Users and Distribution licensee shall fulfil the requirement stipulated in Regulations issued by CEA under sub-section (2) of Section 177 of the Act and amendments issued from time to time while deciding distribution protection system. This chapter describes these minimum standards.

6.2

Objective. - The objective of this chapter is to define the minimum protection requirements for any equipment connected to the distribution system, so that faulty distribution section can be isolated from rest of power system and thereby minimise disruption due to faults.

6.3

General Principles. - (1) No item of electrical equipment shall be allowed to remain connected to the distribution system unless it is covered by appropriate protection aimed at reliability, selectivity, speed and sensitivity of protective relays/devices. Distribution Licensee and users shall co-operate with transmission licensee to ensure correct and appropriate settings of protection to achieve effective, discriminatory removal of faulty equipment within the target clearance time specified in the UERC (State Grid Code) Regulations, 2016, as amended from time to time, (clearance time to be specified from the grid standards or elsewhere)

(2)

Protective relay settings shall not be altered or protection bypassed and/or disconnected without consultation with concerned distribution licensee. In case the protection has been bypassed and/or disconnected by mutual consent, the same should be rectified and protection restored to normal condition as quickly as possible. If no consensus is reached, all the electrical equipments shall be isolated forthwith.

(3)

Ground, Vertical and horizontal clearances of a building structure shall be as specified in relevant provisions of the CEA (Measures relating to safety and electric supply) Regulations, 2010 as amended from time to time.

6.4

Protection Manual. - Distribution Licensee shall prepare and enforce standard manual of protection indicating minimum protection requirement within the distribution system and connected Users' system. The Protection Manual shall cover protection of supply lines and Power and Distribution Transformers through which supply is provided to the consumers. The Protection Manual shall be prepared taking into consideration the UERC (State Grid Code) Regulations, 2016 as amended from time to time and shall contain relevant data on fault levels at various places, guidelines for setting standard relays for over current and earth faults, fuse rating selection criteria, etc. A copy of Protection Manual shall be furnished to Commission in compliance of this requirement after the same is prepared by the Distribution Licensee.

6.5

Protection at inter-connection point of EHT GSS. - All 33 kV and 11 kV lines emanating from EHT GSS shall be provided with a minimum of over current and earth fault protection with or without directional features along with high set element as per the UERC (State Grid Code) Regulations, 2016 as amended from time to time) requirement. Coordination with the originating EHT sub-station should be ensured to avoid major sub-station equipment/ EHT transmission lines from tripping on through faults due to delayed fault clearance in the distribution feeders. Protection on 33 kV and 11 kV transformers and lines (or their sectionalising points) of HT system of Distribution Licensee shall be coordinated with settings of protection provided on 33 kV and 11 kV feeders at EHT sub-stations.

6.6

33 kV and 11 kV line protection. - (1) The settings of protective relays for 33 kV and 11 kV lines from the feeding sub-stations shall be such that a fault in any section does not affect the upstream section between the generating unit/feeding sub-station and the faulty section under all conditions. 33 kV radial lines shall have two over current and one earth fault non-directional IDMT relay protection at feeding station. The relays shall also have instantaneous over current element. Where 33kV line is an interconnection between two sub-stations or a generator unit and the sub-station, these relays shall have directional feature.

(2)

All 33 kV and 11 kV lines at connection points shall be provided with a minimum of over current and earth fault relays as follows:

1

Radial feeders

Non-directional time lag over current and earth fault relays with suitable settings to obtain discrimination between adjacent relays settings.

2

Parallel/ring feeders and interconnected feeders

Directional time lag over current and earth fault relays.

3

Long feeders/transformer feeders

These feeders shall incorporate a high set instantaneous element.

6.7

Transformer Protection. - (1) The transformers shall be suitable for outdoor installation with three phase, 50 Hz, 11 kV or 33 kV system in which the neutral is effectively earthed and they should be suitable for service with fluctuations in supply voltage upto plus 12.5% to minus 12.5%.

(2)

The transformers shall conform to the following specific parameters:

Sl. No.

Item

11 kV Distribution Transformers

33 kV Distribution Transformers

1

System voltage (max.)

12kV

36kV

2

Rated voltage HV

11kV

33kV

3

Rated voltage LV

433-250 V*

433-250 V*

4

Frequency

50 Hz +/- 5%*

50 Hz +/- 5%

5

No. of Phases

Three/Single

Three/Single

6

Connection HV

Delta

Delta

7

Connection LV

Star (Neutral brought out)

Star (Neutral brought out)

8

Vector group

Dyn-11

Dyn-11

9

Type of cooling

ONAN/Dry

ONAN/ONAF/Dry

(3)

The Minimum protection requirements of transformers installed in distribution system shall be as under:

Sl. No.

Type

Primary side Protection

Secondary side Protection

1

Below 1000kVA

Link switch with fuse or circuit breaker of such capacity as to carry the full load current and to break only the magnetising current of transformer

In case of consumer, circuit breaker of adequate rating shall be installed.

In case of generating company or licensee, link switch with fuse or circuit breaker of adequate capacity shall be installed. Buchholz, winding and oil temperature alarm protection

2

1000 kVA and above

Circuit breaker of adequate capacity shall be provided

Circuit breaker of adequate rating shall be provided. Buchholz, winding and oil temperature alarm, and tripping protection.

6.8

Protection Coordination. - (1) The Distribution licensee shall decide the relay settings with the data collected from the

Transmission licensee and the Users on Fault Levels at various EHT sub-stations. Representatives of the generating companies, transmission licensees and distribution licensees shall meet periodically to discuss such malfunctions, changes in the system configuration, if any, and possible revised settings of relays. The Transmission licensee shall notify the initial settings and any subsequent changes to the Distribution Licensee and Users from time to time. Routine checks on the performance of protective relays shall be conducted and any malfunction shall be noted and corrected as soon as possible.

(2)

Transmission Licensee shall be responsible for arranging periodical meetings between the generating companies, transmission licensee and the distribution licensees to discuss coordination of protection as per the UERC (State Grid Code) Regulations, 2016 (as amended from time to time) requirement. The transmission licensee shall investigate any malfunction of protection or other unsatisfactory protection issues. The distribution licensees shall take prompt action to correct any protection malfunction or activity in distribution system as discussed and agreed to in these periodical meetings.

6.9

Clearance front buildings of lines of voltage and service lines not exceeding 650 Volts. - (1) An overhead line shall not cross over an existing building as far as possible and no building shall be constructed under an existing overhead line.

(2)

Where an overhead line of voltage not exceeding 650 V passes above or adjacent to or terminates on any building, the following minimum clearances from any accessible point, on the basis of maximum sag, shall be observed, namely-

(a)

for any flat roof, open balcony, veranda roof and lean-to-roof-

(i)

when the line passes above the building a vertical clearance of 2.5 metres from the highest point, and

(ii)

When the line passes adjacent to the building a horizontal clearance of 1.2 metres from the nearest point, and

(b)

for pitched roof-

(i)

when the line passes above the building a vertical clearance of 2.5 metres immediately under the line, and

(ii)

when the line passes adjacent to the building a horizontal clearance of 1.2 metres.

(3)

Any conductor so situated as to have a clearance less than that specified above shall be adequately insulated and shall be attached at suitable intervals to a bare earthed bearer wire having a breaking strength of not less than 350 kg.

(4)

The horizontal clearance shall be measured when the line is at a maximum deflection from the vertical due to wind pressure.

6.10

Clearances from buildings of lines of voltage exceeding 650 V. - (1) An overhead line shall not cross over an existing building as far as possible and no building shall be constructed under an existing overhead line.

(2)

Where an overhead line of voltage, exceeding 650 V passes above or adjacent to any building or part of a building it shall have on the basis of maximum sag a vertical clearance above the highest part of the building immediately under such line, of not less than-

(i)

for lines of voltages exceeding 650 Volts up to
and including 33,000 Volts

3.7 metres

(ii)

for lines of voltages exceeding 33 kV

3.7 metres plus 0.30 metre for every additional

33,000 Volts or part thereof.

(3)

The horizontal clearance between the nearest conductor and any part of such building shall, on the basis of maximum deflection due to wind pressure, be not less than-

(i)

for lines of voltages exceeding 650 V up to and

including 11,000 Volts

1.2 metres

(ii)

for lines of voltages exceeding 11,000 V and up

to and including 33,000 V

2.0 metres

(iii)

for lines of voltages exceeding 33 kV

2.0 metres plus 0.3 metre for every additional

33 kV or part thereof

Explanation. - For the purpose of Regulations 6.9 and 6.10 the expression "building" shall be deemed to include any structure, whether permanent or temporary.

Chapter 7

Cross Boundary Safety Code

7.

1. Introduction.

- This chapter specifies the requirements for safe working practices for maintenance of equipment associated with cross boundary operations and lays down the procedure to be followed when the work is carried out on electrical equipment connected to another User's System.

Users and Distribution licensee shall also fulfil to the requirements stipulated in Regulations issued by CEA under provisions of clause (b) of sub-section (2) of Section 177 of the Act and amendments issued from time to time while deciding the safety of the distribution system.

7.2

Objective. - The objective of this section is to achieve an agreement on the principles of safety when working across a control boundary between the Distribution Licensee and the Users.

7.3

Control Persons and their Responsibility. - (1) The Distribution Licensee and all the Users (comprising Generating Companies. Transmission Licensees and consumers having load above 1 MVA or dedicated line) shall nominate suitably authorised and technically qualified persons to be responsible for the co-ordination of safety across their boundary in accordance with the CEA (Measures relating to safety and electric supply) Regulations, 2010 as amended from time. These persons shall be referred to as "Control Persons".

(2)

The Distribution Licensee shall issue a list of control persons with their names, designations, addresses and telephone numbers, to all the users having direct control boundary with him. This list shall be updated promptly whenever there is any change of name, designation or telephone/mobile number, e-mail ID of any Control Person named in the list.

(3)

All the Users having a direct control boundary with the Distribution Licensee shall issue a similar list of their Control Persons to the Distribution Licensee. This list shall be updated promptly whenever there is any change of name, designation or telephone/mobile number, e-mail ID of any Control Person named in the list.

(4)

Whenever any work across a cross boundary is to be carried out by the User or the Distribution Licensee, the Control Person of the User or the Distribution Licensee as the case may be, who has to carryout the work, shall directly contact his counterpart. Code words shall be agreed to at the time of work to ensure correct identification of both the parties. Contact between Control Persons shall normally be made by direct telephone.

(5)

If the work extends beyond one shift, the Control Person shall hand over charge to the relief Control Person and fully brief him on the nature of work and the code words in the operation.

(6)

The Control Persons shall co-operate to establish and maintain the precautions necessary to be taken for carrying out the required work in a safe manner. Both the established isolation and the established earth shall be kept in the locked positions wherever such facilities exist, and these shall be clearly identified.

(7)

The Control Person in charge of the work shall satisfy himself that all the safety precautions to be taken are established before commencing the work. He should issue the safety documentation to the working party to allow the work to commence.

(8)

After the completion of the work, the Control Person in charge of the work being carried out should satisfy himself that the safety precautions taken are no longer required, and shall make a direct contact with his counterpart Control Person and request removal of the safety precautions. The equipment shall be declared as suitable for return to service only after confirmation of removal of all the safety precautions, by direct communication, using the code word contact between the two Control Persons, and the return of agreed safety documentation from the working party.

(9)

The Distribution Licensee shall develop an agreed written procedure for Cross Boundary Safety and continuously update the same.

(10)

Any dispute concerning Cross Boundary Safety shall be resolved at the level of STU, if STU is not a party. In case where STU is a party, the dispute shall be referred to the Commission for resolution of the dispute.

7.4

Special Considerations. - (1) All the equipment on Cross Boundary Circuits, which may be used for the purpose of safety co-ordination and establishment of isolation and earthing, shall be permanently and clearly marked with an identification number or name being unique to the particular sub-station. These equipments shall be regularly inspected and maintained in accordance with the manufacturer's specifications.

(2)

Each Control Person shall maintain a legibly written safety log, in chronological order, of all operations and messages relating to the safety co-ordination sent and received by him. All these safety logs shall be retained for a period of not less than five years.

(3)

As far as possible each of the Distribution Licensee shall maintain an updated map of his system pertaining to the area fed by each sub-station. Otherwise the schematic diagram of the system for 11 kV and above shall be maintained and exhibited in the concerned area offices/ feeding sub-stations of the Distribution Licensee.

Chapter 8

Incident/Accident Reporting

8.

1. Introduction.

- This chapter covers procedure of major incident/accident reporting (which occur in Distribution System) by Users to Distribution Licensee and Licensee to Chief Electrical Inspector.

8.2

Major Incident or Accident Reporting. - (1) The Users shall furnish information to the Distribution Licensee regarding any major incident occurring in their system promptly. The Distribution Licensee and the Users shall establish a format and procedure for exchange of information.

(2)

Reporting of accidents shall be in accordance with Intimation of Accidents (Form and Time service of Notice) Rules, 2004, read with Section 161 of The Electricity Act, 2003. If an accident occurs in the distribution system resulting in or likely to have resulted in loss or injury to human or animal life, the Distribution Licensee shall send a telephonic report to the Electrical Inspector within 24 hours of the knowledge of such occurrence. This shall be followed by a report in writing in the form given in Annexure 6 (as per Form A of Intimation of Accidents (Form and Time service of Notice) Rules,

2004), within 48 hours of the knowledge of occurrence of fatal and other accidents.

8.3

Reporting Procedure. - (1) All reportable incidents occurring in the lines and substations in distribution system shall be promptly reported orally by the Licensee whose equipment has experienced the incident, to all other significantly affected Users identified by the Distribution Licensee and the Transmission Licensee. The reporting Distribution Licensee should submit a written report to the Transmission Licensee, in a prescribed format mutually agreed by the Distribution and Transmission Licensee, within one hour of such oral report. If the reporting incident is of major nature, the written report shall be submitted within six hours duly followed by a comprehensive report within 7 days of the submission of the initial written report. In other cases, the reporting Distribution Licensee shall submit a report within fifteen working days to the Transmission Licensee.

(2)

The Transmission Licensee shall call for a report from any Distribution Licensee on any reportable incident affecting other Users and particularly in case such User whose equipment might have been a source of the reportable incident does not report the same. However, this shall not relieve any User from the obligation to report Events in accordance with Rules framed under relevant provision of The Electricity Act, 2003. The format for such a report shall be as per the approval of the Distribution Code Review Panel and shall typically contain the following:

(a)

Location of the incident,

(b)

Date and time of the incident,

(c)

Plant or Equipment involved,

(d)

Supplies interrupted and the duration wherever applicable,

(e)

Amount of Generation loss, wherever applicable,

(f)

System Parameters before and after the incident, (Voltage, Frequency, Load, Generation, etc.)

(g)

Network configuration before the incident,

(h)

Relay indications and performance of protection,

(i)

Brief description of the incident,

(j)

Estimated time of return to service,

(k)

Any other relevant information,

(l)

Recommendations for future improvement,

(m)

Name and designation of the reporting person.

(3)

The report shall contain sufficient detail to describe the event to enable the recipient to assess the implications and risks arising out of the same. The recipient may ask for clarifications and additional information wherever necessary and it is obligatory that the reporting User shall put his best efforts and provide all the necessary and reasonable information.

(4)

In case of a request by either party, the oral report shall be written down by the sender and dictated by way of a telephone/mobile message or sent by Fax/e-mail to the recipient. In case of an emergency the report can be given only orally and followed by written confirmation.

Annexure 1

Load Data for demands of 1 MVA and above to be furnished by the User/Consumer Name and Address of User/Consumer:

Sl. No.

Description

Details

1

Type of Load

(State whether- steel melting furnace loads, Rolling mills, traction loads, other industrial loads, pumping loads, etc.)

2

Maximum Demand (kVA) and Annual Energy

Requirement in kWh

08/03/01

3

Year/Years by which full/part Supply is required

08/03/02

4

Location of Load

(Furnish location map to scale, indicate details of Consumer category/capacity, nearest Railway Station, and nearest EHT sub-station)

5

Rated Voltage at which supply is required.

Whether Single phase or Three-phase supply required

08/03/03

6

Type of supply

Normal/Alternate/Dedicated (specify details)

7

Description of Equipment

08/03/04

A

Motors State purpose and number of installations, voltage and kW rating, starting current, type of motors, types of drives and control arrangements

08/03/05

B

Heating Type and kW Rating

08/03/06

C

Furnace Type, Furnace Transformer Capacity and Voltage Ratio

08/03/07

D

Electrolysis Purpose, kVA capacity

08/03/08

E

Lighting kW Demand

08/03/09

8

Sensitivity of demand to fluctuations in voltage
and frequency of supply at the time of Peak Demand (Give details)

08/03/10

9

Voltage sensitivity

MW/kV MVar/Kv

10

Frequency sensitivity

MW/Hz MVar/Hz

11

Phase unbalance

imposed on system Maximum (%)

Average (%)

08/03/11

12

Maximum harmonic component imposed (Furnish
details of devices included with the system for the suppression
of harmonics, also furnish the harmonic currents of different
orders drawn by each device without filters)

08/03/12

13

Details of any loads, which may cause Demand
fluctuations of greater than 10 MW at the point of connection,
including Voltage Dips (percentage) lasting for 5 seconds and
more. (Give details)

08/03/13

Annexure 2

Embedded Generator Unit-Wise Data

Name and Address of Generating Company:

Location of Generating Plants(s):

Terminal Volts (kV);

Rated kVA;

Maximum and minimum Active Power sent out (kW)

Reactive Power requirements (kVAr), if any;

Type of Generating Plant - synchronous,
asynchronous, etc.;

Fault Level contribution

Method of voltage control;

Generator transformer details, if applicable;

Requirements for Top-up supplies and/or standby
supplies;

Generator kW/kVAr capability chart (at lower
voltage terminals);

Type of excitation system;

Inertia constant kW secs/kVA;

Stator Resistance;

Direct-Axis Reactance (Sub-transient, Transient
and Synchronous);

Quadrature-Axis Reactance (Sub-transient and
Synchronous);

Zero Sequence (Resistance and Reactance);

Negative Sequence (Resistance and Reactance);
Generator Transformer (Resistance, Reactance,
kVA Rating, Tap Arrangement, Vector Group, Grounding, Connection
and % Impedance);

Automatic Voltage Regulator block diagram,
including the data on the gains (forward and feedback), time
constants, and voltage control limits;

Speed Governor block diagram detailing the
governor fly-ball, if applicable, and control system and Prime
Mover time constants, together with the turbine rating and
maximum power

Standby Requirements:

Rated Capacity and Minimum Generation of each
Generating Unit and Power Station in kV for standby capacity
requirements.

Generating Unit and Power Station auxiliary

Demand (Active Power and Reactive Power) in kW and kVAr, at rated
capacity conditions. For Customers Self-Generating Plant, this
shall include Top-up requirements.

Interface Arrangements-the means of
synchronisation between the Distributors and User;

Details of arrangements for connecting to ground
that part of the Generator's System directly connected to the
Distribution System;

The means of connection and Disconnection which
are to be employed

Precautions to be taken to ensure the
continuance of safe conditions should any grounded neutral point
of the Generator's System become disconnected from ground.

Precautions to be taken to ensure the
continuance of safe conditions should any grounded neutral point
of the Generator's System become disconnected from ground.

Annexure 3

System Data to be provided to the intending User/Consumer with contract demand of 1 MVA or more (wherever
applicable)

1. 33 KV and above Distribution Line Data relevant to the location where connection has been applied/ feasible to
provide.
2. Details of metering system and protection system proposed.
3. Fault levels at which the consumer should design his equipment.
4. Fault clearance time for consumer's switch gear and
5. Sub-station fault level.

Annexure 4

Site Responsibility Schedule

Name of Sub-station/Location Site Owner

Name of co-ordination officer of site Telephone No.

Fax No.:

Item of Plant/ Apparatus

Plant owner

Safety Responsibility

Control Responsibility

Operation Responsibility

Maintenance Responsibility

Remarks

.....kV Switchyard

All equipment including bus-bars

Feeders

Generating Units

Other (to be specified)

Signatures

Plant Owner

Safety Responsibility Officer

Control Responsibility Officer

Operation Responsibility Officer

Maintenance Responsibility Officer

Annexure 5

List of Power Factor Apparatus

For Motors

Sl. No.

Rating of Individual Motor

KVAR Rating of Capacitor

750 RPM

1000 RPM

1500 RPM

3000 RPM

1

Upto 3 HP

1

1

1

1

2

5 HP

2

2

2

2

3

7.5 HP

3

3

3

3

3

10 HP

3

3

3

3

5

15 HP

6

5

5
4
6
20 HP 8
7
6
5
7
25 HP
9
8
7
6
8
30 HP
9
9
8
7
9
40 HP
13
11
10
9
10
50 HP
15
15
12
10
11
60 HP
20
20
16
14
12
75 HP
23
23
19
16
13
100 HP
30
30
23
20
14
125 HP

39

38

31

26

15

150 HP

45

45

36

30

16

200 HP

60

60

48

40

For Welding Transformers

Sl. No.

Name Plate Rating in KVA of individual welding
transformer

Capacity of the capacitors (KVAR)

1

1

1

2

2

2

3

3

3

3

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Annexure 6

The Intimation of Accidents (Form and Time of service of Notice) Rules, 2004

Form A

Form for reporting electrical accidents

1. Date and time of accident.

2. Place of accident.

(Village/Town, Tehsil/Thana, District and State).

3. System and voltage of supply (Whether Extra High Voltage (EHV)/High Voltage (HV)/Low Voltage (LV) Line, sub-station/generation station/consumer's installations/service lines/other installations).

4. Designation of the officer-in-charge of the generating company/licensee in whose jurisdiction the accident occurred.

5. Name of owner/user of energy in whose premises the accident occurred.

6. Details of victim(s):

(a)

Human

Sl. No.

Name

Father's Name

Sex of Victim

Full Postal address

Approximate age

Fatal/Non-fatal

1

2

(b)

Animal

Sl. No.

Description of Animal(s)

Number(s)

Name(s) of Owner(s)

Address(es) of owner(s)

Fatal/Non-fatal

1

2

7. In case the victim(s) is/are employee(s) of supplier-

(a)

brief description of the job undertaken, if any;

(b)

whether such person/persons was/were allowed to work on the job.

8. In case the victim(s) is/are employee(s) of a licensed contractor,-

(a)

did the victim(s) possess any electric workmen's permit(s), supervisor's certificate of competency?

If yes, give number and date of issue and the name of issuing authority;

(b)

name and designation of the person, who assigned the duties of the victim(s).

9. In case of accident in the system of the generating company/licensee, was the permit to work (PTW) taken

10. (a) Describe fully the nature and extent of injuries, e.g., fatal/disablement (permanent or temporary) of any portion of the body or burns or other injuries.

(b)

In case of fatal accident, was the post mortem performed?

11. Detailed causes leading to the accident.

(To be given in a separate sheet annexed to this form).

12. Action taken regarding first aid, medical attendance, etc., immediately after the occurrence of the accident (give details).

13.

Whether the District Magistrate and Police Station concerned have been informed of the accident (if so, give details).

14.

Steps taken to preserve the evidence in connection with the accident to extent possible.

15.

Name and designation(s) of the person(s) assisting, supervising the person(s) killed or injured.

16. What safety equipments were given to or used by the person(s) who met with this accident (e.g. rubber gloves, rubber mats, safety bells and ladders, etc.)?

17.

Whether isolating switches and other sectionalising devices were employed to deaden the sections for working on the same? Whether working section was earthed at the site of work?

18.

Whether the work on the live lines was undertaken by authorised person(s)? If so, the name and the designation of such person(s) may be given.

19.

Whether artificial resuscitation treatment was given to the person(s) who met with the electric accident? If yes, how long was it continued before its abandonment?

20.

Names and designations of persons present at, and witnessed, the accident.

21.

Any other information/remarks.

Signature

Place:Time:Date:

NameDesignationAddress of the person reporting

Annexure 7

Check List for Reports/Plans/Procedures

Submitted by

Submitted to

Agency

SLDC

STU/TL

Consumer

Distribution Licensee

Generator

DCRP

Commission

Electrical Inspector

Consumer

3.3(2), 3.15(2), 5.3(2), 5.10(1) and (6), 7.3(3),

8.2(1)

2.7(1)

2.7(1)

Distribution Licence

5.4(5) (c), 5.5(3) (c) and (e)

3.4(1), (4) and (5), 3.5(1), 5.4(1), 8.3(1) and
(2)

3.15(2), 5.4(6), 5.10(1), (2) and (9), 7.2(2)

3.15(2), 5.10(1), (2) and (9), 7.3(2)

1.5(4), 3.4(5), 3.14(2) (a), 3.16(4), 5.6(2),

5.15(4), 5.20, 6.4

8.2(2)

Generator

3.15(2), 5.3(2), 5.10(1). 7.3(3)

DCRP

1.8, 2.3(2)

Commission

2.3(1), 2.7(1) and (2)

Related AI tags, queries and research notes

Translation